

Food Volume & Calorie Estimation

VORGESTELLT VON

Lukas Lommer & Polina Kochmar

MODELL & PROJEKT

Clemens Dieterich , Jochen Lichtenscheidel , Nico Gergen

Food Volume & Calorie Estimation



?

How much food is on the plate?

?

How many grams?

?

How many calories?



Estimate food amount and calories from plate images



Volume | Grams | Kcal estimation

Input Data & Challenge

Reliable estimation depends on image perspective and available visual information.



Top Image

Visible: food area on the plate — shape and footprint of each item.



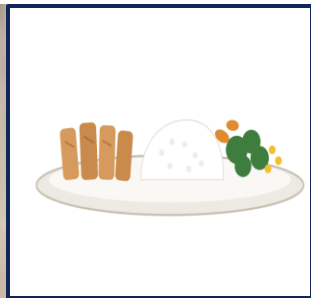
Side Image

Visible: food height and profile — key input for real 3D volume.



Without a side image: no reliable real volume — only a grams-based fallback is possible.

Top + Side



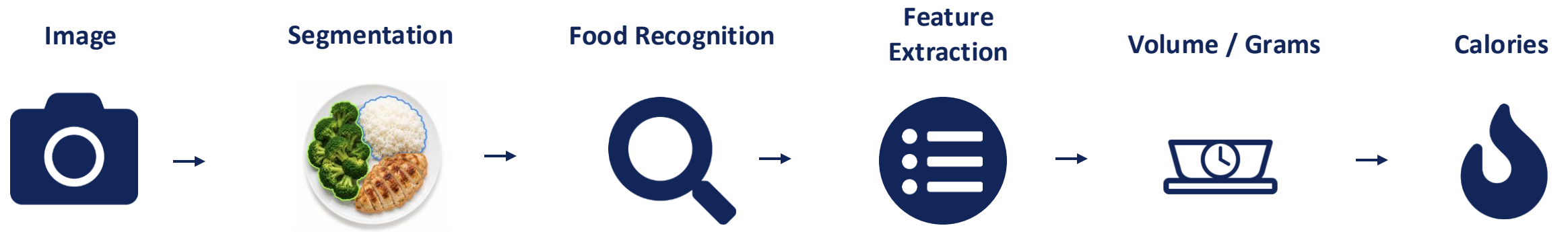
Real volume estimation (ml)

Only Top



Grams fallback — no real volume

Images are transformed into model-ready features before estimation.



Pipeline Components



FastSAM

Segments individual food items on the plate.



CLIP

Recognizes / filters food classes by semantic similarity.



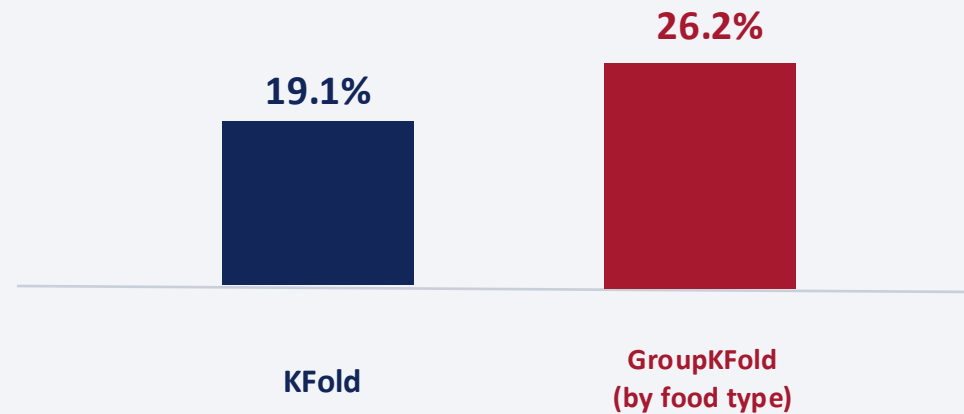
Gradient Boosting Regressor

Volume fallback using area_cm2, height_cm, area_x_height.

The model works, but performance depends strongly on image quality and viewpoint.

Evaluation

MAPE by validation strategy



Limitations



Photo quality



Segmentation errors



Scale estimation



No reliable volume without side image